



Hydrochemical Modelling of Seawater Intrusion and Geogenic Salinity for Sustainable Groundwater Management in Coastal Aquifers

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Abstract: Groundwater in coastal aquifers is highly vulnerable to salinisation processes driven by both seawater intrusion and geogenic sources. Understanding these processes is essential for developing sustainable groundwater management strategies. This study presents a hydrochemical modelling approach to identify and quantify the main processes controlling groundwater composition in a coastal aquifer. The methodology integrates physicochemical parameters and ionic composition data to simulate mixing scenarios between freshwater, seawater, and geogenic sources using the pH-REdox-Equilibrium in C language software (PHREEQC). The results indicate that salinity in coastal wells is primarily controlled by seawater intrusion, while inland areas are significantly influenced by interactions with evaporitic and carbonate basement formations. Transitional zones exhibit mixed hydrochemical signatures, reflecting the combined influence of these processes. These findings provide a process-based framework to support groundwater management decisions, including pumping regulation, well rotation, and managed recharge strategies. The proposed approach contributes to improving water security and long-term sustainability in coastal aquifer systems.

Keywords: Groundwater; Hydrochemical modelling; pH-REdox-Equilibrium in C language software; Manglaralto river-aquifer system; Water quality

1. Introduction

Access to safe water remains a major challenge for sustainable development in developing countries, particularly in Latin America (Gomez et al., 2019), where approximately 25% of the population lacks access to this resource (CEPAL, 2023). The unequal distribution of water resources, coupled with insufficient infrastructure and increasing pressure from climate change, exacerbates the difficulties in providing drinking water to communities. According to UNESCO data (United Nations, 2024; WWAP United Nations World Water Assessment Programme, 2016), watersheds in these regions are often subject to increasing productive activities and population growth, leading to a significant rise in water demand that frequently exceeds sustainable limits.

In coastal areas, where surface water sources are scarce or seasonal, aquifers become the primary source of supply (Bocanegra et al., 2010; Custodio, 2010; Montoya et al., 2019). However, these systems are highly

vulnerable to saltwater intrusion due to their proximity to the sea, a vulnerability exacerbated by intensive exploitation and changes in water dynamics caused by climate and human activities (Datta et al., 2009; Hussain et al., 2019; Mastrocicco & Colombani, 2021). Without proper management, the encroachment of seawater into the aquifer can irreversibly compromise the quantity and quality of water, forcing communities to resort to costly treatments or complex infrastructure to ensure water supply (Hamdan et al., 2021; Saad et al., 2023). In severe cases, water levels can drop to the point that the aquifer becomes unusable. In this context, two key questions guide the present work: (i) how do seawater intrusion and geogenic processes control the hydrochemical evolution of groundwater in coastal aquifers? and (ii) how can this process-based understanding inform sustainable groundwater management strategies under increasing anthropogenic and climatic pressures? Addressing these questions is essential for translating hydrochemical diagnosis into actionable measures for community-based water security, which constitutes the central aim of this study.

Climate change poses a constant threat to coastal aquifers due to their vulnerability to saltwater intrusion. Rising global temperatures and changes in precipitation and evapotranspiration patterns affect natural recharge, reducing the aquifer's renewal capacity (Dao et al., 2024). Furthermore, rising sea levels facilitate the inland migration of saltwater, degrading the quality of stored freshwater (Pham et al., 2023; Richardson et al., 2024). This phenomenon creates challenges for water sustainability and compromises water availability for human consumption, agricultural irrigation, and industrial activities, thereby increasing conflicts over water use.

Currently, several strategies exist to improve groundwater management in coastal areas, including artificial aquifer recharge (Bonilla Valverde et al., 2018; Javadi et al., 2015; Lal & Datta, 2018; Mekni & Souissi, 2016; Motallebian et al., 2019), e.g., Water Sowing and Harvesting (WS&H) (Herrera-Franco et al., 2024), optimization of groundwater use through controlled pumping strategies (Huang & Chiu, 2018; Yang et al., 2021), and the adoption of desalination technologies (Idrees, 2020; Schroeder et al., 2021; Takabatake et al., 2021). In parallel, continuous monitoring through hydrochemical and geophysical studies enables assessment of water quantity and quality and mitigation of risks associated with saltwater intrusion (El-Sayed et al., 2023; Kumar et al., 2023; Morante-Carballo et al., 2026; Ukpai & Okogbue, 2017). These measures are essential to guarantee the water security of coastal communities and preserve associated ecosystems.

In Manglaralto, located in Ecuador's coastal region, the exploitation of the alluvial aquifer is crucial to meet the water demands of the population and of economic activities such as tourism, gastronomy, and agriculture. However, factors such as aquifer overexploitation, increased water demand due to the growth of the local and seasonal population, the lack of strict management regulations, and the effects of climate change on rainfall have intensified saltwater intrusion and decreased piezometric levels, compromising the sustainability of the water resource in this area and jeopardising water supply (Malavé-Hernández & Carrión-Mero, 2025; Sánchez-Zambrano et al., 2025). This scenario poses a concrete management dilemma: how can a community water board operating under limited technical and financial resources prioritise interventions when the causes of salinisation may vary substantially from one well to another within the same aquifer? Answering this question requires a process-based hydrochemical diagnosis capable of distinguishing between seawater intrusion and geogenic salinisation sources, so that management actions such as pumping regulation, well rotation, and managed recharge can be targeted where they are most effective.

The Manglaralto coastal aquifer has been the subject of several hydrogeological studies due to its socioeconomic importance at the community level. Previous research has characterised its hydrochemical and geophysical properties, identifying freshwater and brackish-water facies, as well as local and lateral mixing and infiltration processes (Carrión-Mero et al., 2021a). However, significant gaps remain in understanding flow dynamics, seasonal effects, and the interactions between human activities and natural processes, which could help improve understanding and management within a sustainability framework. The lack of comprehensive hydrochemical models in the Manglaralto River basin limits the ability to predict and manage water quality processes in the aquifer. Therefore, the implementation of advanced hydrochemical simulations (Kempka et al., 2022; Ryskie et al., 2024), such as those enabled by the pH-REdox-Equilibrium in C language software (PHREEQC) (Parkhurst & Appelo, 1999), represents a key step towards sustainable aquifer management by providing a detailed view of solute transport and reactions within the aquifer system.

The use of hydrochemical models, such as those provided by PHREEQC, allows for the simulation of complex interactions between surface water and groundwater, accounting for the solutes present in the aquifer, chemical kinetics, reactive transport, and chemical equilibrium phases (Manoj et al., 2019; Muniruzzaman & Rolle, 2016). These models are instrumental in coastal systems, where saltwater intrusion dynamics are influenced by interactions among tides, groundwater recharge, and human activities (Petalas & Lambrakis, 2006; Selvakumar et al., 2022).

To address the research questions posed above, this study develops a hydrochemical model of the Manglaralto coastal aquifer by integrating ionic content data and physicochemical parameters from seven monitoring wells. The specific objectives are: (i) to identify and quantify the dominant salinisation processes (seawater intrusion versus geogenic contributions); (ii) to simulate the hydrochemical evolution of the aquifer under dry and wet season conditions; and (iii) to derive management-relevant insights that support adaptive decision-making by the

local water management board Junta Administradora de Agua Potable de Manglaralto (JAAPMAN). Beyond its technical contribution, the study aims to strengthen the scientific basis for community-based water security in Latin American coastal rural communities, where groundwater sustainability is increasingly threatened by overexploitation, climate variability, and the absence of formal management frameworks.

1.1 Geographical and Climatic Context

The Manglaralto coastal aquifer is located on the southwest coast of Ecuador, north of the province of Santa Elena, within the Manglaralto hydrological basin (Figure 1a). This aquifer is managed by the Manglaralto Regional Drinking Water Management Board (JAAPMAN, by its acronym in Spanish) using WS&H techniques, which include a traditional aquifer recharge system (a technical-artisanal dam) and 15 wells (Figure 1b). The aquifer serves as the water supply for six rural communities in the area (Libertador Bolívar, San Antonio, Río Chico, Cadeate, Manglaralto, and Montañita), encompassing approximately 40,000 inhabitants, including both permanent and seasonal populations (Maldonado et al., 2019).

The region has a semi-arid climate, with an average annual temperature of 24 °C. Rainfall follows a unimodal pattern, with a marked dry season from May to November (~10 mm/month) and a wet season beginning in December, with rainfall peaking in February-March (~100 mm/month) (Ramírez, 2023).

1.2 Geological Context

The geological context encompasses five main units: (i) volcano-sedimentary rocks of the Cayo Formation (Upper Cretaceous) that make up the aquifer basement; (ii) lithological units of the Ancon Group (Eocene), such as reef limestones and carbonate sandstones of the Javita Limestone Member (Socorro Formation); conglomerates, siltstones, shales, sandstones, and clays of the Dos Mangas Member (Socorro Formation); and sandstones, siltstones, shales, conglomerates, and limestones of the Seca Formation, which together comprise the impermeable layer that favors water trapping; (iii) chocolate-brown sandstones and clays with gypsum veinlets, and conglomeratic sandstones of the Tosagua Formation (Oligocene-Miocene); (iv) conglomerates and sands of the Tablazo Formation (Pleistocene-Holocene); and (v) Quaternary (Holocene) alluvial deposits (gravel, sand, clay, and silt) that make up the alluvial aquifer (Figure 1b). The Manglaralto aquifer covers approximately 508 ha and has an estimated water volume of 13.6 hm³.

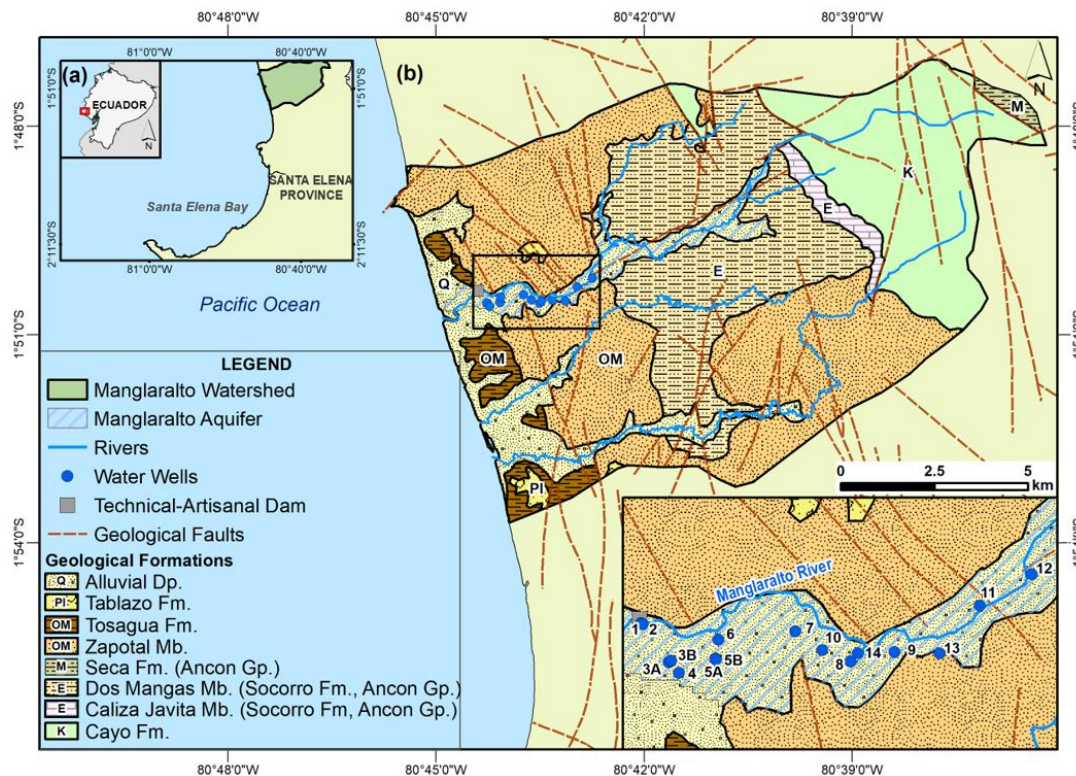


Figure 1. Map of the study area. (a) Location of the Manglaralto basin with respect to the Equator, (b) Manglaralto Basin and associated lithology

1.3 Aquifer Hydrochemistry and Vulnerability

Previous isotopic evidence indicates that the Manglaralto aquifer waters are recharged mainly through local flows with short residence times, which increases the system's sensitivity to solute dilution or concentration processes, depending on the availability of meteoric water (Carrión-Mero et al., 2021a).

From a hydrochemical perspective, groundwater samples from the Manglaralto aquifer exhibit calcium-chloride facies in the wells closest to the coastline (W2, W3, and W4), which evolve towards sodium-bicarbonate facies during the rainy season due to meteoric recharge. In wells farther from the coast, both calcium- and sodium-bicarbonate facies are observed. This evolution is supported by analysis of the Hydrochemical Facies Evolution Diagram (HFE-D) (Carrión-Mero et al., 2021a), which allows for the identification of both recharge processes and marine intrusion events.

In the context of climate change, with events such as La Niña intensifying periods of low rainfall, the vulnerability of the Manglaralto aquifer is further accentuated by additional factors. These include high levels of evapotranspiration linked to climatic conditions; aquifer overexploitation; and point-source discharge of untreated wastewater and septic tank effluent, especially during periods of lower dilution, which increases the risk of both saltwater intrusion and groundwater contamination (Toala et al., 2020; Ramírez, 2023).

2. Methodology

The methodology involved hydrochemical modelling of the waters of the Manglaralto coastal aquifer. The open-source software PHREEQC developed by the United States Geological Survey (USGS) was used to simulate cation-exchange reactions and reconstruct water-mixing processes to determine the extent of saltwater intrusion in the community water wells. The methodological approach comprised three phases: (i) parameter selection, (ii) modelling with PHREEQC, and (iii) model evaluation (Figure 2).

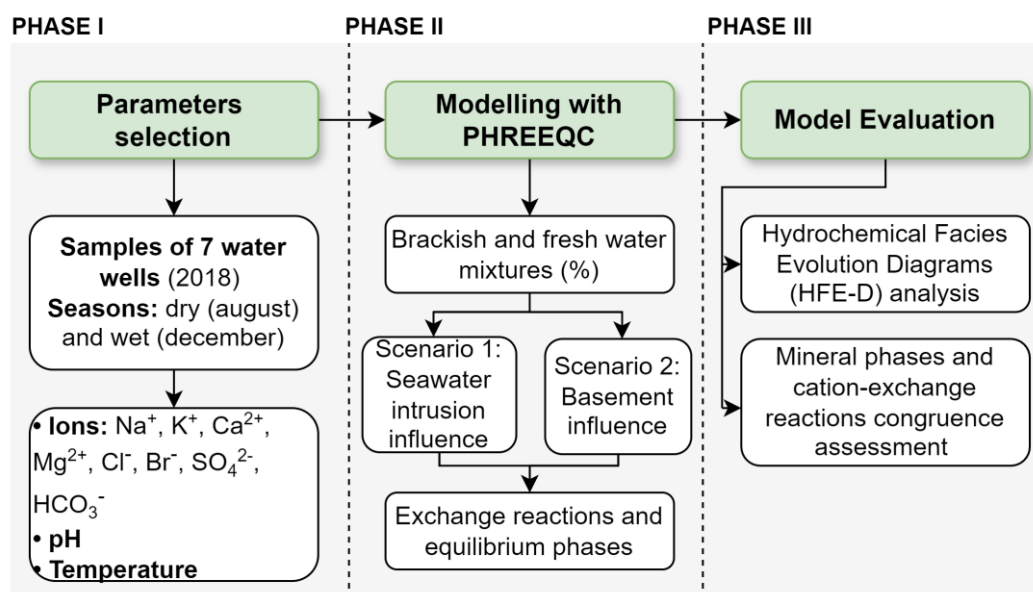


Figure 2. Methodological scheme of the study

2.1 Phase I: Selection of Hydrochemical Parameters

The modelling data included parameters from the hydrochemical characterisation of the Manglaralto aquifer, conducted at seven water wells (W2, W3, W4, W6, W10, W11, and W12) during the dry (August) and wet (December) seasons of 2018. The ionic contents of Na⁺, K⁺, Ca²⁺, Mg²⁺, Cl⁻, Br⁻, SO₄²⁻, and HCO₃⁻ were included, as well as the physicochemical parameters pH and temperature, due to their direct impact on solute solubility and the ion exchange capacity to form compounds. Table 1 summarises the selected parameters for hydrochemical modelling.

The seven wells analysed in this study (W2, W3, W4, W6, W10, W11, and W12) were selected from the wells operated by JAAPMAN at the time of the 2018 sampling campaigns. They are distributed along the main groundwater flow corridor of the Manglaralto aquifer, which follows the course of the Manglaralto River for approximately 3 km from the technical-artisanal recharge dam, located near the coastline, towards the inland sector dominated by the basement formations. The proximal wells (W2, W3, and W4) are located within the first

kilometre downstream of the dam and represent the sector most directly exposed to the saltwater intrusion front, where the hydrochemical response to the saline wedge is therefore strongest. The intermediate well W6 lies in the transitional zone, where the influence of seawater becomes progressively diluted, and mineral interactions with the basement begin to gain importance. Finally, the distal wells W10, W11, and W12 are located further inland, where the hydrochemical signature is dominated by the dissolution of evaporitic and carbonate basement minerals rather than by marine inputs. This spatial distribution captures the gradient of hydrochemical processes governing the aquifer, ranging from seawater-influenced to geogenically influenced wells, while reflecting the dynamic interaction and mixing between freshwater, brackish, and basement-derived waters along the flow path.

The remaining wells operated by JAAPMAN were not included in the present analysis because they were constructed after the 2018 sampling campaigns to meet the increasing water demand of the communities served by the aquifer. Their siting was guided by the hydrogeological conditions of the basin, prioritising zones with greater saturated thickness to ensure long-term productivity. Because these wells currently experience higher pumping pressure than the original supply network, future research should incorporate them into the monitoring framework to assess how the redistribution of extraction stress modifies the aquifer's hydrochemical response over time.

The 2018 dataset corresponds to a hydrologically near-neutral year in ENSO terms and therefore represents the typical seasonal behavior of the aquifer rather than extreme hydroclimatic conditions. This makes the dataset suitable for characterizing the baseline hydrochemical processes controlling salinization, while the assessment of extreme events (El Niño/La Niña years) is identified as a priority direction for future research. Previous hydrochemical, isotopic, and geophysical studies in the same wells (Carrión-Mero et al., 2021a; Ramírez, 2023; Sánchez-Zambrano et al., 2025) have reported consistent salinization patterns, reinforcing the representativeness of the selected dataset for process-based modelling.

Table 1. Selected hydrochemical parameters for modelling

Parameter	Season	W2	W3	W4	W6	W10	W11	W12
pH	DS	7.1	7.2	7.7	7.5	7.2	7.2	7.5
	WS	7.5	7.5	7.2	7.2	7.4	7.4	7.8
T (°C)	DS	26.5	26.9	26.0	26.1	24.5	24.5	25.2
	WS	27.3	26.7	26.4	25.6	26.4	25.9	24.4
Na ⁺ (mg/L)	DS	274.0	187.0	317.0	126.0	110.0	74.0	93.0
	WS	176.0	141.0	216.0	96.0	127.0	75.0	84.0
K ⁺ (mg/L)	DS	18.0	12.0	14.0	8.1	6.4	7.4	8.2
	WS	11.0	5.4	8.6	6.0	5.4	6.5	6.4
Ca ²⁺ (mg/L)	DS	588.0	239.0	346.0	96.0	76.0	87.0	90.0
	WS	294.0	62.0	167.0	73.0	70.0	78.0	67.0
Mg ²⁺ (mg/L)	DS	110.0	46.0	73.0	18.0	16.0	15.0	20.0
	WS	55.0	16.0	35.0	14.0	14.0	14.0	16.0
Cl ⁻ (mg/L)	DS	1224.0	370.0	723.0	90.0	108.0	86.0	101.0
	WS	487.0	80.0	81.0	70.0	82.0	71.0	78.0
Br ⁻ (mg/L)	DS	5.1	2.1	3.8	0.2	0.6	0.4	0.4
	WS	1.6	0.3	1.1	0.2	0.3	0.2	0.3
SO ₄ ²⁻ (mg/L)	DS	272.0	176.0	394.0	178.0	137.0	121.0	149.0
	WS	170.0	118.0	240.0	1138.0	123.0	104.0	108.0
HCO ₃ ⁻ (mg/L)	DS	303.0	309.0	383.0	327.0	323.0	303.0	330.0
	WS	293.0	280.0	340.0	323.0	327.0	269.0	277.0

Note: W: well, DS: dry season (August), WS: wet season (December).

2.2 Phase II: Modelling with pH-REdox-Equilibrium in C Language Software

To simulate the geochemical processes affecting the hydrochemical evolution of the Manglaralto coastal aquifer, PHREEQC software version 3.7.3 was used. The modelling focused on the period of the highest ionic content (the dry season). Progressive mixing between brackish water and the meteoric water characteristic of the rainy season was modelled to determine the evolution towards reference conditions. The water-mixing modelling considered two scenarios associated with the main sources of salinity affecting the aquifer: seawater intrusion into wells closest to the shoreline and interaction with the geological basement in distal wells.

In wells W2, W3, W4, and W6, mixing scenarios between seawater and freshwater were simulated, starting with a 50%–50% ratio, and subsequently adjusting the proportions until reaching the dilutions that predicted the concentrations observed during the rainy season (35% in W2, 40% in W3, 20% in W4, and 10% in W6). The selection of proportions was based on the relative distance to the coast and the direction of groundwater flow defined in previous transport studies (Carrión-Mero et al., 2021b). The concentrations used for seawater are presented in Table 2.

The mixing ratios were determined through a stepwise calibration procedure that integrated hydrogeological

constraints with quantitative fitting against field data. The initial proportions for each well were derived from the groundwater flow directions and hydraulic gradients reported in the flow and transport model of Carrión-Mero et al. (2021b), which established that wells closer to the shoreline receive a higher contribution from the saline wedge than those located further inland. Using these values as the starting point, the proportions were refined iteratively in 5% increments until the deviation between simulated and observed concentrations was minimised against the dry-season reference, which represents the period of maximum ionic content in the aquifer.

Table 2. Concentrations in the seawater parameters used for the model

Parameters	Units	Value
pH	-	8.22
pE	-	8.45
ρ	g/cm ³	1.02
T	°C	25.00
Ca ²⁺	mg/L	412.30
Mg ²⁺	mg/L	1291.80
K ⁺	mg/L	399.10
Si	mg/L	4.28
Cl ⁻	mg/L	19353.00
Alkalinity	mg/L de HCO ₃ ⁻	141.68
SO ₄ ²⁻	mg/L	2712.00
Br ⁻	mg/L	65.00

In wells with less direct marine influence, equilibrium reactions with specific minerals and cation exchange processes were considered. Equilibrium phases with calcite and dolomite were defined based on bicarbonate enrichment and the Na-Cl to Mg-HCO₃ facies transition, while typical basement mineral phases—including gypsum, halite, and sylvite—were incorporated depending on the ionic deficits observed in each well. The active cation exchangers were Ca²⁺, Mg²⁺, and Na⁺, with weights adjusted according to the cationic differences between dry and wet seasons. Table 3 summarises the cation exchange weights applied in each well, and Table 4 details the mineral phases included in the modelling, indicating in each case whether they were implemented through the EQUILIBRIUM_PHASES keyword (allowing dissolution or precipitation until thermodynamic equilibrium) or through the REACTION keyword (i.e., adding a fixed molar amount independently of the saturation state).

Table 3. Cation exchange weights applied in the pH-REdox-Equilibrium in C language software (PHREEQC) modelling for each well (W)

Parameters	Exchangers		
	Ca ²⁺	Mg ²⁺	Na ⁺
W2	-	-	-
W3	0.60	0.20	0.60
W4	0.30	0.20	0.50
W6	-	-	-
W10	-	-	-
W11	-	-	-
W12	-	-	-

Note: Values represent the molar weights assigned to each cation exchanger (CaX₂, MgX₂, NaX) in the PHREEQC input files. A dash (-) indicates that cation exchange reactions were not applied for that well.

Table 4. Mineral phases included in the pH-REdox-Equilibrium in C language software (PHREEQC) modelling for each well (W), with their corresponding molar weights

Well	Calcite	Dolomite	Gypsum	Halite	Sylvite
W2	-	-	-	-	-
W3	0.1 ^a	0.1 ^a	-	-	-
W4	0.0 ^a	0.0 ^a	-	-	-
W6	-	-	0.0009 ^b	-	-
W10	-	-	0.0003 ^b	0.0003 ^b	0.0001 ^b
W11	-	0.0001 ^b	0.0004 ^b	0.0005 ^b	0.0001 ^b
W12	-	0.0001 ^b	0.0004 ^b	0.0005 ^b	0.0001 ^b

Note: A dash (-) indicates that the phase was not included in the model for that well.

Values represent the molar weights assigned in the PHREEQC input files. Superscripts indicate the PHREEQC keyword used: ^a Implemented through the EQUILIBRIUM_PHASES keyword (i.e., the phase is allowed to dissolve or precipitate until thermodynamic equilibrium, saturation indices (SI) $\approx$ 0, is reached). Phases declared

with a weight of 0.0 (W4) were defined in the input file but kept inactive in the present simulation. ^b Implemented through the REACTION keyword (i.e., a fixed molar amount is added to the solution regardless of the saturation state).

The influence of the basement was specifically simulated in wells W6, W10, W11, and W12. In well W6, the combined calcium and sulfate deficiency was reproduced by including an equilibrium phase with gypsum. In well W10, halite, sylvite, and gypsum were used to compensate for the Na⁺, Cl⁻, K⁺, Ca²⁺, and SO₄²⁻ deficits, while in wells W11 and W12, dolomite was also incorporated to adjust carbonate levels. These reactions were designed considering an evaporite sequence compatible with the aquifer geology.

Mixing scenarios between brackish water (from the sea or the basement) and meteoric water were also modelled with proportions ranging from 65%–35% to 95%–5% (brackish–fresh). The PHREEQC results, expressed in molalities, were converted to mg/L using conversion factors based on molecular weights and ionic proportions for each species.

Within the framework of hydrochemical modelling, SI were calculated to assess the equilibrium state between groundwater and the mineral phases potentially present in the aquifer. The SI calculation is based on Eq. (1):

$$SI = \log \left(\frac{IAP}{K_{sp}} \right) \quad (1)$$

where, IAP corresponds to the ionic activity product of the species that constitute the mineral, and K_{sp} represents the solubility constant of the mineral at the simulation temperature (Parkhurst & Appelo, 1999; Parkhurst & Appelo, 2013).

An SI = 0 value indicates equilibrium between the solution and the mineral; SI > 0 values indicate supersaturation conditions that favour precipitation; and SI < 0 values show unsaturation conditions, with a tendency toward dissolution. PHREEQC estimates ionic activities using activity coefficient models based on the findings of Banks et al. (1931) and Hückel & Debye (1923), which allows for an accurate diagnosis of the dissolution-precipitation processes in the modelled aquifer system.

The calibration was anchored by the conservative ions Cl⁻ and Br⁻, which are not significantly affected by precipitation–dissolution or cation-exchange reactions and therefore constrain the extent of mixing between freshwater and brackish endmembers. Once the mixing proportions were defined, the reactive ions (Na⁺, K⁺, Ca²⁺, Mg²⁺) were adjusted through the cation-exchange and mineral-equilibrium reactions described above, while sulfate and bicarbonate concentrations were controlled by the equilibrium phases involving gypsum, calcite, and dolomite. The quality of the overall fit was quantified using the mean absolute percentage error (MAPE) between simulated and observed concentrations for the eight major ions (Na⁺, K⁺, Ca²⁺, Mg²⁺, Cl⁻, Br⁻, SO₄²⁻, HCO₃⁻) across the seven modelled wells (Table 5). The calibrated ratios produced MAPE values of approximately 10–27% for the major reactive ions and the conservative Cl⁻, all within the uncertainty range reported in conceptual hydrochemical modelling studies of coastal aquifers where full inverse modelling is not applied (Gopinath et al., 2016; Petalas & Lambrakis, 2006). The higher MAPE obtained for Br⁻ (51%) reflects the sensitivity of the metric to the very low absolute Br⁻ concentrations measured in the inland wells (typically below 0.6 mg/L), where small absolute deviations translate into large relative values; it does not indicate a poor reproduction of the geochemical behaviour of this ion. Detailed observed and simulated values for all wells and ions are provided as Supplementary Material (Table A1). This procedure ensures that the reported mixing proportions are hydrogeologically grounded and quantitatively justified across the full ionic composition of the aquifer, rather than empirically adjusted on a single tracer.

Table 5. Absolute percentage error (APE, %) between simulated and observed concentrations of major ions at the optimal mixing proportion for each well, and overall MAPE per ion across the seven modelled wells (W)

Well	Mixture	Na ⁺	K ⁺	Ca ²⁺	Mg ²⁺	Cl ⁻	Br ⁻	SO ₄ ²⁻	HCO ₃ ⁻
W2	95–5%	92.5	32.8	50.1	12.9	8.9	27.2	6.9	22.5
W3	95–5%	2.5	0.5	12.9	63.8	22.3	11.2	3.1	19.8
W4	85–15%	1.5	15.5	30.3	51.7	16.3	22.2	26.8	5.3
W6	95–5%	16.7	4.1	12.1	11.1	85.4	158.7	30.2	16.6
W10	85–15%	4.5	32.8	5.8	25.6	22.3	54.6	2.6	12.6
W11	85–15%	1.7	27.5	0.7	4.5	5.1	46.8	4.9	2.8
W12	90–10%	6.3	17.9	10.7	15.8	9.4	39.2	0.1	2.1
MAPE	(all wells)	18.0	18.7	17.5	26.5	24.2	51.4	10.7	11.7

Note: Values represent the absolute percentage error ($APE = 100 \times |C_{obs} - C_{sim}| / C_{obs}$) at the optimal mixing proportion. MAPE = mean absolute percentage error across the seven wells. Higher MAPE for Br⁻ reflects the metric's sensitivity to very low absolute concentrations (typically below 0.6 mg/L in inland wells), where small absolute deviations produce large relative errors.

2.3 Phase III: Model Evaluation

Because the dry-season hydrochemical observations were used as the reference for the iterative calibration described in Section 2.2, the present study does not include an independent predictive validation step. Instead, the model evaluation was based on two semi-independent consistency checks that rely on diagnostic indicators not used in the calibration procedure itself.

The first consistency check used HFE-D to verify that the simulated evolution of the samples reproduced the field-observed facies transitions, particularly between saline-intrusion facies in the proximal wells and meteoric-recharge or basement-dissolution facies in the distal wells. Within the calibration procedure, the conservative ions Cl^- and Br^- played a key role in defining the extent of mixing between freshwater and brackish endmembers, since their behaviour is not significantly affected by precipitation–dissolution or cation-exchange reactions; consequently, their simulated concentrations reflect almost exclusively the imposed mixing proportions. The HFE-D analysis complements this by evaluating the geochemical trajectory of the system as a whole rather than the absolute concentrations of individual ions, and therefore provides a conceptual check independent of the MAPE metric used in the calibration.

The second consistency check assessed the congruence between the mineral phases and cation-exchange reactions implemented in PHREEQC and the expected compositions of the basement lithofacies. The presence of evaporite minerals (gypsum, halite, sylvite), carbonates (calcite, dolomite), and the cation exchangers (Ca^{2+} , Mg^{2+} , Na^+) was contrasted with the geological characterisation of the Manglaralto basin and with patterns reported for similar coastal aquifers in carbonate and evaporite environments. The agreement between the modelled equilibrium phases and the geological setting reinforces the physical plausibility of the simulated processes.

It should be emphasised that the present approach corresponds to a conceptual hydrogeochemical model rather than to a fully independently validated predictive model. Its purpose is to provide a process-based diagnosis of the dominant salinisation mechanisms (seawater intrusion versus geogenic contributions) and to support adaptive management decisions, while acknowledging that fully independent predictive validation would require multi-year monitoring datasets and complementary geophysical and isotopic information; these are identified as priority directions for future work.

3. Results

Numerical modelling was applied to seven monitoring wells (W2, W3, W4, W6, W10, W11, and W12) distributed along the coastal aquifer. The concentrations obtained using the model were compared with the values measured during the dry season, selecting the ratio that showed the best match. Additionally, saturation indices were calculated for the main mineral phases in the selected mixture for each well. The model considered two scenarios: the first for wells proximal to the coast, where marine brackish water exerts a predominant influence, and the second for wells that exhibit simultaneous recharge by meteoric waters and dissolution of mineral species precipitated from the basement.

3.1 Scenario 1: Wells Influenced by Brackish Marine Water

The first numerical modelling scenario considered wells W2, W3, and W4 and simulated mixtures of brackish water of marine origin (due to seawater intrusion) and continental freshwater. The seawater source was progressively diluted to reproduce the intrusion conditions in this sector of the aquifer, based on the location of the wells and the transport model proposed by Carrión-Mero et al. (2021b). Reactions with the basement rock were not considered in these wells because of the high seawater concentration in the transition zone. For each well, sequential mixtures of the diluted solution and seawater were applied at 65–95% dilution.

In well W2 (Figure 3a), an optimal mixture of 90–95% groundwater with 5–10% diluted seawater was identified. The mixture exhibited supersaturation in minerals such as aragonite, calcite, and dolomite, while minerals such as halite, sylvite, gypsum, and quartz remained unsaturated, promoting their dissolution (Figure 4a).

For well W3 (Figure 3b), an initial dilution of 40% was used, with a final mixture estimated at 95% groundwater and 5% seawater. Equilibrium reactions with calcite and dolomite (weights of 0.1 each) and cation exchange reactions for Mg^{2+} (0.2), Na^+ (0.6), and Ca^{2+} (0.6) were modelled. This modelling indicates supersaturation of dolomite and calcite, and a tendency towards unsaturation of most of the remaining mineral species (Figure 4b).

In well W4 (Figure 3c), an initial dilution weight of 80% was applied. The simulations achieved the target concentrations for most species; however, slight deficiencies in calcium and sulfate were detected. Equilibrium phases included calcite and dolomite, in addition to exchange reactions for Ca^{2+} , Mg^{2+} , and Na^+ . Furthermore, the mixtures did not exhibit significant supersaturation (Figure 4c). These results indicate that wells located closer to the coastline are highly vulnerable to salinisation driven by seawater intrusion, suggesting that these zones require priority management actions such as controlled pumping and monitoring.

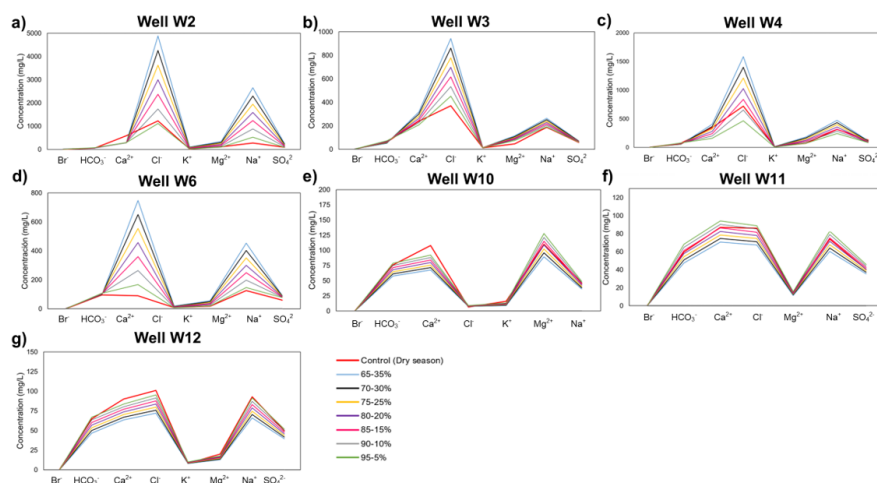


Figure 3. Mixture modelling for the study of wells between salinity sources and recharging conditions
Note: Bromide ion (Br^-), Bicarbonate ion (HCO_3^-), Calcium ion (Ca^{2+}), Chloride ion (Cl^-), Magnesium ion (Mg^{2+}), Sodium ion (Na^+), Sulfate ion (SO_4^{2-}).

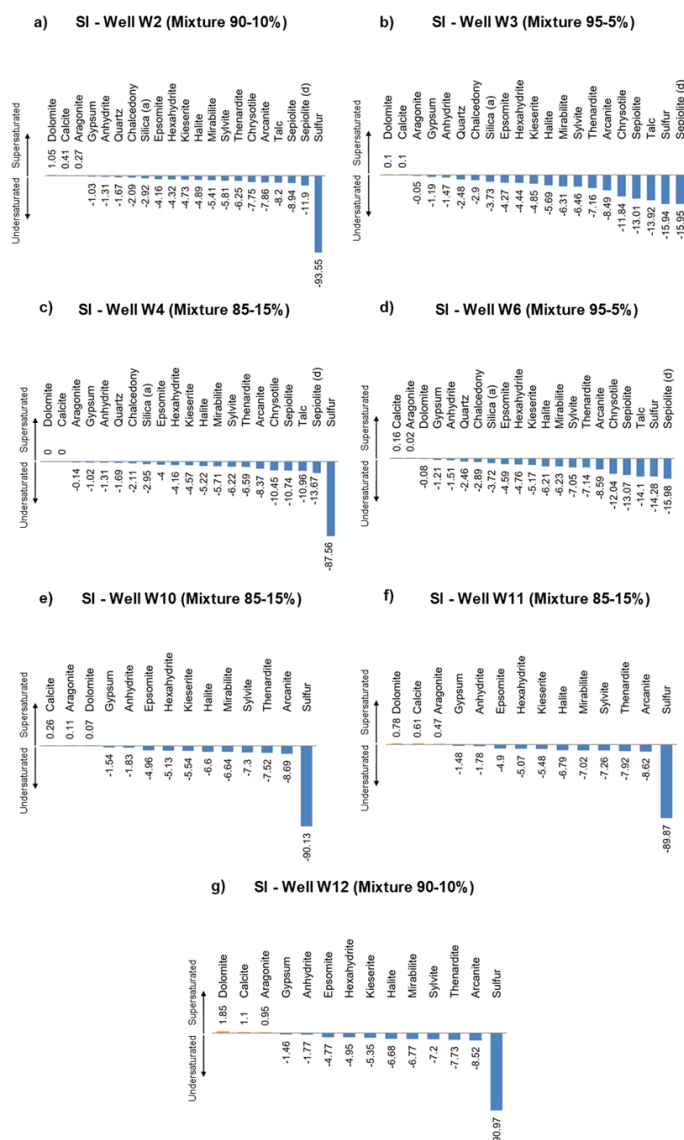


Figure 4. Saturation indices (SI) of the study wells
Note: amorphous (a), disordered (d).

3.2 Scenario 2: Wells Influenced by Meteoric Water and Basement Water

The second numerical modelling scenario considered W6, W10, W11, and W12. In these wells, the salinity source is controlled by the dissolution of basement minerals. Only in well W6 is a partial influence of dilute seawater recognised, and an initial dilution to 10% was performed. The modelled concentrations for well W6 are shown in Figure 3d. A reaction with gypsum was applied to all mixtures, using a molar weighting factor of 0.0009. The 65–35% mixture exhibited supersaturation, leading to talc precipitation. Unsaturated conditions were recorded in the remaining mixtures (Figure 4d). The reaction models were performed between 14 and 15 iterations.

The second scenario modelled the distal wells W10, W11, and W12, whose hydrochemical composition reflects a greater interaction with the geological basement. The water in these wells showed concentrations that cannot be explained solely by dilution; therefore, equilibrium phases with evaporite minerals such as gypsum, halite, sylvite, and dolomite were integrated. These models were adjusted based on characteristic ionic deficits and the facies types observed in the samples.

In well W10, the modelling included halite, sylvite, and gypsum to adjust the concentrations of Na^+ , Cl^- , K^+ , Ca^{2+} , and SO_4^{2-} (Figure 3e). The best match with the observed values was obtained for the 85–15% mixture, in which the carbonate phases (calcite, aragonite, and dolomite) showed positive saturation indices. The remaining mixtures exhibited a gradient of carbonate supersaturation: calcite alone for mixtures below 70%, calcite and aragonite between 75–80%, and calcite, aragonite, and dolomite for mixtures above 80% (Figure 4e).

In the modelling of wells W11 (Figure 3f) and W12 (Figure 3g), dolomite was added to adjust the carbonate concentrations. The models confirmed that the waters reflect basement mineral dissolution processes, with low supersaturation indices, and a geochemical dynamic conditioned by mixing with short-residence meteoric waters that infiltrate and reach the aquifer.

In well W11, halite, sylvite, gypsum, and dolomite were incorporated as reactive phases. The 85–15% ratio most accurately reproduced the observed concentrations. In this mixture, aragonite, calcite, and dolomite exhibited supersaturation, while the remaining phases evaluated remained unsaturated (Figure 4f). The molar weights were adjusted iteratively until the most representative model reproducing the expected concentrations was found: 0.0005 for halite, 0.0001 for sylvite, 0.0004 for gypsum, and 0.0001 for dolomite.

Finally, in well W12, the most representative mixtures corresponded to proportions between 90–10% and 95–5%. Within this range, aragonite and calcite were supersaturated, while dolomite reached supersaturation in mixtures exceeding 90%. The other mineral phases were unsaturated (Figure 4g). The molar weights were adjusted iteratively until the most representative model reproducing the expected concentrations was found: 0.0005 for halite, 0.0001 for sylvite, 0.0004 for gypsum, and 0.0001 for dolomite.

Table 6 summarises the main conditions and results for the wells in each modelling scenario.

In contrast to the proximal wells, the inland wells (W10, W11, and W12) are primarily influenced by geogenic salinity derived from the basement, indicating that management strategies in these areas should focus on monitoring mineral dissolution processes and maintaining balanced extraction rates.

Table 6. Conditions and results for modelling scenarios

Well	Mixture Type	Optimal Proportion (%)	Minerals in Equilibrium	Cation Exchange	Supersaturated Phases (SI > 0)	Undersaturated Phases (SI < 0)
W2	Diluted seawater	90–95/10–5	Aragonite, calcite, dolomite	Not applied	Dolomite (1.05), Calcite (0.41), Aragonite (0.27)	Gypsum, halite, epsomite, kieserite, sylvite, thenardite, arcanite
W3	Diluted seawater	95/5	Calcite, dolomite	Na, Ca, Mg	Calcite (0.10), Dolomite (0.10)	Aragonite, talc, halite, epsomite, sepiolite, thenardite
W4	Diluted seawater	85/15	Calcite, dolomite	Ca, Mg, Na	Calcite (0.00), Dolomite (0.00)	Aragonite, talc, gypsum, halite, sepiolite, thenardite
W6	Diluted seawater	90–95/10–5 (estimated)	Calcite, aragonite	Not applied	Calcite (0.16), Aragonite (0.02)	Dolomite, halite, talc, gypsum, sepiolite, thenardite
W10	Meteoric + basement	-	Halite, sylvite, gypsum, dolomite	Not applied	Dolomite (0.23), Calcite (0.33), Aragonite (0.19)	Gypsum, halite, thenardite, arcanite
W11	Meteoric + basement	-	Halite, sylvite, gypsum, dolomite	Not applied	Dolomite (0.83), Calcite (0.64), Aragonite (0.49)	Gypsum, halite, thenardite, arcanite
W12	Meteoric + basement	-	Halite, sylvite, gypsum, dolomite	Not applied	Dolomite (1.86), Calcite (1.10), Aragonite (0.96)	Gypsum, halite, thenardite, arcanite

4. Discussion

The hydrochemical model, considering the ionic contents of Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^- , Br^- , SO_4^{2-} , HCO_3^- , as well as the physicochemical parameters pH and temperature in seven water wells of the Manglaralto aquifer, allowed

the recognition of a hydric geochemistry related to processes of confluence of meteoric waters, seawater and solutions of the aquifer basement.

The simulation data indicated that the concentrations recorded during the dry season in the Manglaralto aquifer wells are reproducible using a model of progressive water mixtures in the basin. The interaction between continental groundwater with bicarbonate facies and marine water with chloride facies had to be adjusted using dissolution or precipitation reactions of carbonates and evaporite minerals, as well as cation-exchange reactions, where the hydrochemical conditions required it. Adjusting the solutions in 5% intervals allowed visualisation of the hydrochemical transition between the dry and wet seasons and identification of imbalances between measured and simulated contents. This behaviour demonstrates that the hydrochemical response is influenced by dilute solutions that depend on physical conditions, primarily hydraulic gradients controlled by pumping, meteoric recharge conditions, and basement dissolution (Chidambaram et al., 2012; Petalas & Lambrakis, 2006).

The analysis indicates the combined influence of two salinity sources on the aquifer. The first source concerns the crystalline basement and Pleistocene marine deposits on the southern coast of Ecuador, which are the products of marine transgression and regression events on the Santa Elena Peninsula. These events generated marine terraces and favoured the retention of residual salts and connate waters trapped in the sedimentary formations and basement fractures (Pedoja et al., 2006a; Pedoja et al., 2006b). This behaviour is consistent with geological studies, which indicate that the accreted basement is covered by mixed (marine-continental) lithofacies deposits with evaporite content, which influences the incidence of gypsum, halite, and undersaturated or supersaturated carbonates in the simulations (Pedoja et al., 2006a; Valencia Robles, 2017).

The second source of salinity is seawater intrusion, which occurs naturally in coastal environments and can be intensified by the overexploitation of groundwater resources. In Manglaralto, the growing demand for water for human consumption and tourism has led to piezometric declines and an inverted hydraulic gradient, promoting the advance of the brackish water wedge into the aquifer (Ramírez, 2023). This trend is consistent with what has been observed in other semi-arid coastal aquifers (Gopinath et al., 2016), where inadequate exploitation leads to accelerated salinisation and increases vulnerability under normal dry conditions and during prolonged dry periods due to the effects of climate change. These findings confirm that anthropogenic pressure, together with climate variability, is the predominant factor determining water quality evolution, and that the developed model enables the management of groundwater exploitation thresholds, accounting for the effects of seawater intrusion under low rainfall conditions.

In the Manglaralto aquifer, the basement rock plays a crucial role in determining the hydrochemical facies. On the one hand, it provides a large amount of minerals to the water, and on the other, it controls the hydrogeochemical processes of mineral dissolution by altering the saturation state of carbonate and evaporite phases. The positive saturation indices for carbonate species such as calcite, aragonite, and dolomite in wells W2, W4, and W6 suggest a contribution from the local lithology, which mixes with chloride facies ions from the seawater. Other mineral phases, such as gypsum, halite, and potassium salts, remained unsaturated and would be susceptible to dissolution in the wells within the basin. This behaviour is consistent with that reported in coastal aquifers of Asia and Europe, where the freshwater-saltwater interface generates geochemical instability and localised dissolution (Chidambaram et al., 2012; Gopinath et al., 2016; Telahigue et al., 2018).

The implementation of dilutions and mixtures in the model was necessary to reproduce the system's actual hydrochemical conditions. The aquifer waters result from complex interactions among seasonal meteoric recharge, seawater intrusion, and salt contributions from the basement. The approximation using successive mixtures adequately and progressively reflects the intrusion and evolution between the dry and wet seasons, consistent with the methodology applied in other coastal aquifers (Petalas & Lambrakis, 2006; Telahigue et al., 2018). For example, the high sodium values and calcium deficiency in well W2 suggest the possibility of cation exchange reactions, although the HFE-D model did not show significant variability in Na-Ca. In the case of well W3, the limitations in the system's neutralisation capacity against saline influences were associated with excess chlorides and magnesium, indicating that the mixture equilibrium was maintained without the need to increase the proportion of seawater.

The model has limitations that must be considered when interpreting the results. Hydrogeological heterogeneity and anisotropy are only partially represented, so processes such as preferential flow, transverse dispersion, and diffusion in low-permeability zones are not explicitly included. Likewise, biogeochemical processes such as bacterial sulfate reduction, which can modify the geochemistry in anoxic zones (Garing et al., 2013), were not considered due to data limitations; these processes are even considered as one of the hypotheses explaining the deficiency in sulfate content in well W4, along with possible reductions in organic material (Gomis-Yagües et al., 2000). Finally, the timescale was restricted to the dry-wet transition, leaving unassessed the influence of interannual oscillations and extreme events, such as El Niño, which can modify intrusion and recharge dynamics. Similarly, the use of a single hydrological year (2018), which corresponds to near-neutral ENSO conditions, limits the ability to capture the aquifer response under extreme wet (El Niño) or extreme dry (La Niña) scenarios; a multi-year monitoring dataset would be required to fully characterise the sensitivity of the hydrochemical processes to interannual climatic variability. Therefore, future research can incorporate these analyses to obtain a more

comprehensive view.

This study represents a significant step forward in the management of the Manglaralto aquifer. By identifying the most representative mixing ranges and associated saturation indices, tools are generated to establish strategies that prevent overexploitation and control the advance of the saltwater wedge. Measures derived from the findings include planned well pumping rotation, regulation of pumping rates during the dry season, and the continuation of controlled aquifer recharge programs implemented by the local water management board (JAAPMAN) using alluvial dams that mitigate saline intrusion during periods of peak demand or through other ways of artificial recharge (e.g., recharge ponds (Abd-Elaty et al., 2021)). Furthermore, continuous monitoring of hydrochemical parameters and periodic model updates would allow adaptation of management strategies to critical scenarios arising from climate variability, population growth, and the addition of new users to the distribution network.

The results support the need to limit exploitation under conditions that increase the aquifer's vulnerability, implement well-pumping rotation to reduce the pressure generated by water abstraction, and even delve deeper into managed recharge processes, such as alluvial dams (Dillon et al., 2020). Furthermore, they highlight the need to monitor hydrochemical parameters and to review the temporal evolution of models for the wet and dry seasons (Samani, 2024; Stevanović & Stevanović, 2021). Ultimately, understanding the Manglaralto aquifer as a complex system of reactions between the basement and water mixtures enables appropriate and sustainable management of the resource despite increasing anthropogenic pressure and climate change.

The integration of hydrochemical modelling with groundwater management provides a framework for linking process understanding with sustainability outcomes. The identification of zones dominated by seawater intrusion and geogenic salinity allows the prioritisation of management actions, including pumping regulation, well rotation, and managed aquifer recharge.

From a governance perspective, these results can support decision-making by local water management institutions, such as JAAPMAN, by providing a scientific basis for adaptive management strategies. The model can be used as a decision-support tool to evaluate the impacts of extraction scenarios and climate variability, contributing to long-term water security and system resilience in coastal communities.

The model results can be translated into well-specific management strategies that build on the operational practices currently implemented by JAAPMAN. The board currently performs monthly multiparametric physicochemical in situ monitoring (e.g., electrical conductivity, salinity, resistivity, and Total Dissolved Solids), and rotates the operational wells based on salinity, using the more saline wells (W2 and W4) preferentially during periods of higher demand and mixing their water with that of less saline wells to maintain delivery quality. This practice is consistent with the hydrochemical zonation identified by the model and provides a solid baseline for implementing the following refinements.

First, the proximal wells (W2, W3, and W4), which are dominated by seawater intrusion, should remain under the most intensive monitoring effort. During the dry season (May–November), when piezometric levels and meteoric recharge are at their lowest, increasing the frequency of in situ conductivity and salinity measurements (e.g., from monthly to biweekly) would enable earlier detection of saline-front advances. The current strategy of preferentially activating W2 and W4 only under high demand is consistent with the model, but should be paired with predefined salinity thresholds at which their operation is reduced or temporarily suspended in favour of less saline wells.

Second, the transitional well W6 should be treated as an early-warning point: any progressive increase in Cl^- or salinity in this well would indicate an inland advance of the saline front, warranting an immediate reassessment of the extraction regime in the proximal sector. Monthly multiparameter monitoring is appropriate, with laboratory follow-up if anomalies are detected.

Third, the distal wells (W10, W11, and W12), which are controlled by geogenic salinisation, do not require the same intensity of conductivity monitoring, but their biannual laboratory analyses should explicitly include SO_4^{2-} , HCO_3^- , and Ca^{2+} as indicators of mineral dissolution processes from the basement. Sustained increases in these solutes would signal accelerated weathering or longer residence times, both relevant for long-term planning of extraction rates. These wells can also serve as preferred sources for mixing with the more saline proximal wells, as JAAPMAN already does, to maintain delivery quality during high-demand periods.

Finally, the modelling results suggest that the periods of greatest hydrochemical vulnerability coincide with the late dry season (September–November), when both intrusion and basement dissolution processes are at their peak. Concentrating monitoring efforts and management decisions around this window would optimise the board's limited operational resources. Table 7 summarises the proposed monitoring and management framework, intended as an indicative reference adaptable to JAAPMAN's operational capacity.

Taken together, the findings of this study outline a coherent problem–solution–impact pathway for groundwater sustainability in the Manglaralto aquifer. The problem—progressive salinisation under combined pressure from seawater intrusion, geogenic contributions, overexploitation, and climate variability—is diagnosed through hydrochemical modelling at the individual well scale. The solution emerges from the spatial zonation identified by the model, which enables the targeted application of management strategies: controlled pumping and monitoring in coastal wells dominated by seawater intrusion, and regulated extraction with dissolution monitoring

in inland wells controlled by geogenic processes. The impact, in turn, lies in transferring these insights to JAAPMAN as an accessible and replicable decision-support tool, transforming technical modelling outputs into actionable governance practices that reinforce the long-term water security of the communities served by the aquifer.

Table 7. Indicative monitoring and management framework for the Manglaralto coastal aquifer based on the hydrochemical zonation identified by the model and the practices currently implemented by JAAPMAN

Well Group	Dominant Process	Key Parameters to Monitor	In-Situ Monitoring Frequency (Dry Season)	In-Situ Monitoring Frequency (Wet Season)	Laboratory Analyses	Recommended Management
Proximal (W2, W3, W4)	Seawater intrusion	electrical conductivity, salinity, Cl^-	Biweekly	Monthly	Twice a year, full ion suite	Activate W2 and W4 only under high demand, with predefined salinity thresholds for suspension; favour mixing with distal wells
Transitional (W6)	Mixed (intrusion + basement)	electrical conductivity, salinity, Cl^- , SO_4^{2-}	Monthly	Monthly	Twice a year, full ion suite	Use as early-warning point; trigger reassessment of proximal sector if Cl^- increases progressively
Distal (W10, W11, W12)	Geogenic (basement dissolution)	electrical conductivity, SO_4^{2-} , HCO_3^- , Ca^{2+}	Monthly	Quarterly	Twice a year, with emphasis on SO_4^{2-} , HCO_3^- , Ca^{2+} trends	Maintain steady extraction; serve as preferred source for mixing with proximal wells to dilute salinity during high-demand periods

Note: This framework is intended as an indicative reference and is fully adaptable to the operational, technical, and financial capacity of JAAPMAN. Frequencies should be interpreted as minimum recommendations under normal hydroclimatic conditions; under prolonged dry periods or extreme events (La Niña), the in-situ monitoring frequency in the proximal wells should be intensified accordingly; JAAPMAN = Junta Administradora de Agua Potable de Manglaralto.

5. Conclusions

This study reveals the hydrochemical dynamics of groundwater in the Manglaralto aquifer. Ionic contents and physicochemical data from samples taken from seven wells constitute the hydrochemical model. The successive mixing approach replicates the aquifer's hydrochemical conditions, reflecting complex interactions among seasonal meteoric water recharge, seawater intrusion, and basement salt dissolution. Wells W2, W3, and W4 are affected by seawater intrusion, while wells W6, W10, W11, and W12 are affected by basement salt dissolution. Analysis of saturation indices showed that carbonates (calcite, aragonite, and dolomite) are supersaturated in all wells, driven by the direct influence of the carbonate basement. Phases such as gypsum, halite, and potassium salts remain unsaturated, reflecting the influence of the basement and pumping-induced gradients on the hydrochemical evolution of the aquifer.

The research acknowledges the aquifer's hydrochemical interactions, which are essential for management that respects the natural physical environment. In this regard, actions such as continuous monitoring, controlled extraction rates, well-pumping rotation, and managed recharge are key to planning community water sustainability. Simulated scenarios demonstrate that limiting extractions during dry periods and improving recharge structures, such as alluvial dams, can reduce the advance of the brackish wedge and the aquifer's vulnerability to intrusion, even under conditions of high-water demand and low rainfall.

Future research directions could focus on refining the model by incorporating more complex geochemical processes (e.g., mineral dissolution kinetics, redox reactions) and isotopic modelling. Identifying the aquifer's recharge and discharge zones is key to improving model interpretation, predicting the effects of intensive exploitation, and designing artificial recharge strategies. Integrating hydrochemical, isotopic, and piezometric monitoring data from long-term campaigns will enable evaluation of the influence of extreme events such as El Niño and La Niña, thereby strengthening the community-based water sustainability approach in Manglaralto.

From a sustainability perspective, this study illustrates how a conceptual hydrochemical model, even when based on a limited dataset, can inform management actions that support water security in coastal rural communities. By identifying which wells are dominated by seawater intrusion and which are controlled by geogenic processes, the model provides JAAPMAN with a baseline, process-based decision-support framework to refine its current practices of controlled pumping, well rotation, and water mixing, primarily under non-extreme hydroclimatic

conditions. Extending this framework to extreme scenarios (El Niño/La Niña years) and validating it with multi-year monitoring data are identified as priority directions for future work. With these caveats, integrating hydrochemical diagnosis with community-based governance offers a transferable approach to improving groundwater management in similar coastal aquifer systems across Latin America, particularly in contexts where scientific tools must be accessible to local water boards operating under limited institutional resources.

Author Contributions

Conceptualization, F.J.M., J.A.D.-A., and P.C.-M.; methodology, F.J.M., J.A.D.-A., and P.C.-M.; software, F.J.M. and J.A.D.-A.; validation, F.J.M. and J.A.D.-A.; formal analysis, F.J.M. and J.A.D.-A.; investigation, F.J.M., J.A.D.-A., J.M.-H., and P.C.-M.; resources, F.J.M. and J.A.D.-A.; data curation, F.J.M. and J.A.D.-A.; writing—original draft preparation, F.J.M., J.A.D.-A., J.M.-H., and P.C.-M.; writing—review and editing, F.J.M., J.A.D.-A., J.M.-H., and P.C.-M.; visualization, F.J.M., J.A.D.-A., and J.M.-H.; supervision, P.C.-M.; project administration, P.C.-M.; funding acquisition, P.C.-M. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

UNESCO	United Nations Educational, Scientific and Cultural Organization
JAAPMAN	Manglaralto Regional Drinking Water Management Board
HFE-D	Hydrochemical Facies Evolution Diagram
USGS	United States Geological Survey
pH	dimensionless hydrogen potential
T	temperature, °C
Na^+	sodium ion, mg/L
K^+	potassium ion, mg/L
Ca^{2+}	calcium ion, mg/L
Mg^{2+}	magnesium ion, mg/L
Cl^-	chloride ion, mg/L
Br^-	bromide ion, mg/L
SO_4^{2-}	sulfate ion, mg/L
HCO_3^-	bicarbonate ion, mg/L
pE	dimensionless oxidation-reduction potential
Si	silicium, mg/L
IAP	Ionic Activity Product
K_{sp}	solubility product constant
PHREEQC	pH-REdox-Equilibrium in C language software
APE	absolute percentage error
SI	saturation indices

Greek symbols

ρ density, g/cm³

Appendix

The following table presents the complete set of observed concentrations (dry season, August 2018), simulated

concentrations from the PHREEQC model at the optimal mixing proportion identified for each well, and the corresponding absolute percentage error (APE) for the eight major ions (Na⁺, K⁺, Ca²⁺, Mg²⁺, Cl⁻, Br⁻, SO₄²⁻, HCO₃⁻). The bottom row reports the mean absolute percentage error (MAPE) per ion, averaged across the seven wells. Concentrations are given in mg/L.

APE was computed as $APE = 100 \times |C_{obs} - C_{sim}| / C_{obs}$. MAPE corresponds to the arithmetic mean of APE values across the seven wells for each ion. The higher MAPE obtained for Br⁻ (51.4%) reflects the sensitivity of relative error metrics to very low absolute concentrations, with measured Br⁻ values typically below 0.6 mg/L in the inland wells (W10, W11, W12), where small absolute deviations translate into large relative values. Excluding Br⁻, the global MAPE across all reactive and conservative ions ranges between 10.7% (SO₄²⁻) and 26.5% (Mg²⁺), which is consistent with the uncertainty range reported in conceptual hydrochemical modelling of coastal aquifers where full inverse modelling is not applied (Gopinath et al., 2016; Petalas & Lambrakis, 2006).

Table A1. Observed, simulated, and absolute percentage error values for all major ions across the seven modelled wells of the Manglalaralto coastal aquifer

Well	Mixture	Parameter (mg/L)	Na ⁺	K ⁺	Ca ²⁺	Mg ²⁺	Cl ⁻	Br ⁻	SO ₄ ²⁻	HCO ₃ ⁻
W2	95–5%	Observed	274.00	18.00	588.00	110.00	1224.00	5.10	272.00	303.00
		Simulated	527.39	23.90	293.59	95.79	1115.35	3.71	253.12	371.18
		APE (%)	92.5	32.8	50.1	12.9	8.9	27.2	6.9	22.5
W3	95–5%	Observed	187.00	12.00	239.00	46.00	370.00	2.10	176.00	309.00
		Simulated	191.62	12.06	208.14	75.33	452.38	2.33	181.46	370.33
		APE (%)	2.5	0.5	12.9	63.8	22.3	11.2	3.1	19.8
W4	85–15%	Observed	317.00	14.00	346.00	73.00	723.00	3.80	394.00	383.00
		Simulated	321.86	11.84	241.32	110.74	840.95	2.96	288.56	362.76
		APE (%)	1.5	15.5	30.3	51.7	16.3	22.2	26.8	5.3
W6	95–5%	Observed	126.00	8.10	96.00	18.00	90.00	0.20	178.00	327.00
		Simulated	147.07	7.77	107.61	19.99	166.84	0.52	231.70	381.25
		APE (%)	16.7	4.1	12.1	11.1	85.4	158.7	30.2	16.6
W10	85–15%	Observed	110.00	6.40	76.00	16.00	108.00	0.60	137.00	323.00
		Simulated	114.93	8.50	71.58	11.90	83.92	0.27	133.43	363.74
		APE (%)	4.5	32.8	5.8	25.6	22.3	54.6	2.6	12.6
W11	85–15%	Observed	74.00	7.40	87.00	15.00	86.00	0.40	121.00	303.00
		Simulated	75.29	9.44	86.37	14.33	81.65	0.21	126.90	311.57
		APE (%)	1.7	27.5	0.7	4.5	5.1	46.8	4.9	2.8
W12	90–10%	Observed	93.00	8.20	90.00	20.00	101.00	0.40	149.00	330.00
		Simulated	87.16	9.67	80.36	16.83	91.50	0.24	149.18	323.10
		APE (%)	6.3	17.9	10.7	15.8	9.4	39.2	0.1	2.1
MAPE	(all wells)	(%)	18.0	18.7	17.5	26.5	24.2	51.4	10.7	11.7

Note: APE = Absolute Percentage Error; MAPE = Mean Absolute Percentage Error; W = Well.

(i) The optimal mixing proportion (column “Mixture”) corresponds to the brackish-to-meteoric ratio that minimized the deviation between simulated and observed concentrations during the iterative calibration described in Section 2.2 of the manuscript. (ii) The cells in light grey contain the observed and simulated values, while the cells highlighted in pale yellow contain the absolute percentage error (APE). The bottom row, highlighted in blue, reports the MAPE per ion across the seven wells. (iii) SO₄²⁻ and HCO₃⁻ concentrations were derived from PHREEQC outputs for elemental S and C, respectively, using the standard stoichiometric ratios (SO₄ = S × 96.06/32.07; HCO₃ = C × 61.02/12.01). (iv) Higher APE values observed for some ions in individual wells (e.g., Mg²⁺ in W3 and W4, Na⁺ and Ca²⁺ in W2) are discussed in Section 4 of the manuscript in the context of model limitations, particularly the partial representation of hydrogeological heterogeneity and biogeochemical processes (e.g., sulfate reduction) not explicitly included in the present conceptual model.